

Industrial Electrical Infrastructure

A Definitive Guide to Conductor Selection, Sizing, and Installation

UCL Engineering Reference

September 18, 2026

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Chapter 1

Executive Summary and Regulatory Framework

1.1 Introduction

The design, installation, and verification of electrical systems within industrial environments constitute a discipline that sits at the intersection of materials science, thermal thermodynamics, and rigorous safety legislation.

This guide serves as a comprehensive reference for the "competent person" designated to ensure system integrity. It specifically addresses the department's preference for flexible cabling (H07RN-F) in fixed installations where permissible, while providing the necessary requirements for Steel Wire Armoured (SWA) cables where mechanical protection demands it.

1.2 The Legal and Safety Context

The statutory basis for all electrical work in the United Kingdom is the Electricity at Work Regulations 1989 (EAWR), which mandates that all systems must be constructed and maintained to prevent danger. BS 7671:2018+A2:2022 (The IET Wiring Regulations) serves as the non-statutory standard which, if followed, is likely to satisfy the requirements of the EAWR.

1.2.1 The Electricity at Work Regulations 1989 (EAWR)

The EAWR forms the primary statutory basis for all electrical work in the United Kingdom. It is legally binding and mandates that all electrical systems must be constructed, installed, and maintained in a manner that prevents danger. For industrial infrastructure at UCL, this specifically requires that:

- Systems are robust enough to withstand the fault currents and environmental conditions they are subjected to.
- Every person designing, installing, or testing the infrastructure possesses the appropriate technical knowledge and experience (competency).
- Effective means are provided to isolate the system from the supply in an emergency.

1.2.2 BS 7671:2018+A2:2022 (The IET Wiring Regulations)

While the EAWR sets the legal requirement to prevent danger, BS 7671 provides the non-statutory technical framework to achieve it. Compliance with BS 7671 is strongly accepted as satisfying the requirements of the EAWR. In the context of this guide, BS 7671 dictates:

- **Protection against Electric Shock:** Primarily achieved through Automatic Disconnection of Supply (ADS), requiring strict coordination between cable sizing and protective devices.
- **Protection against Thermal Effects:** Ensuring cables do not pose a fire risk due to overload, utilizing precise environmental derating factors.
- **Verification:** Mandating the initial testing protocols (continuity, insulation resistance, and earth loop impedance) detailed in Chapter 8.

1.2.3 Provision and Use of Work Equipment Regulations 1998 (PUWER)

PUWER legally mandates that all equipment provided for use at work is suitable for its intended purpose, safe to use, and maintained in a safe condition. For electrical infrastructure, this means:

- Cables and connectors must be specified for the exact environmental stresses they will face (e.g., vibration, chemical exposure).
- Mandatory verification testing (such as Z_s and Insulation Resistance) forms the basis of the legal maintenance requirement.
- Operators must have immediate access to safety instructions and operating limitations (facilitated via QR codes and the Electrical Competence App).

1.2.4 Dangerous Substances and Explosive Atmospheres Regulations 2002 (DSEAR)

DSEAR requires employers to control the risks of fire and explosions. In propulsion and power electronics labs, this is heavily triggered by battery off-gassing (lithium-ion thermal runaway) and hydrogen fuel cell testing. For cable installations, DSEAR dictates that:

- Standard IP-rated glands are insufficient in zoned explosive atmospheres; ATEX-certified (Ex d or Ex e) barrier glands must be specified.
- Cable routes must not act as conduits for explosive gases to travel between containment zones.

1.2.5 Supply of Machinery (Safety) Regulations 2008

These regulations apply to bespoke machinery built in-house, such as custom dynamometers, high-speed motor test-beds, and automated actuators. From an infrastructure perspective, this mandates that:

- Electrical control circuits must be fail-safe and integrated with physical mechanical guards.
- Hardwired Emergency Stop (E-Stop) infrastructure must physically disconnect motive power to actuators, overriding all software commands.
- Cable routing must account for the mechanical movement of the machinery without causing fatigue or snagging hazards.

The core objectives are the safety of persons and property against:

- Electric shock (basic and fault protection).
- Thermal effects (fire and burns).
- Overcurrent (overload and fault current).

1.3 Scope of Analysis

This document guides the engineer through the lifecycle of an electrical circuit design:

1. **Conductor Selection:** Prioritising Class 5 flexible conductors and EPR insulation.
2. **Sizing and Calculations:** Application of the adiabatic equation, voltage drop limitations, and environmental correction factors (C_a , C_g , C_i).
3. **Installation:** Methodologies for deploying H07RN-F and SWA, including clipping distances and support.
4. **Termination:** Gland selection, ferrule application for Class 5 conductors, and industrial plug specification (BS EN 60309).
5. **Verification:** Testing protocols required to validate the design.

Chapter 2

Conductor Science and Classification

The selection of an electrical cable begins with the conductor itself. We prioritize flexibility for ease of manipulation and vibration resistance.

2.1 The Physics of Conductor Classes

2.1.1 Class 1: Solid Conductors

Consists of a single, solid wire.

- **Properties:** High mechanical rigidity.
- **Application:** Domestic fixed wiring (Twin and Earth).
- **Restriction:** Strictly prohibited for connections to vibrating machinery or mobile equipment due to the risk of work-hardening and fatigue fracture.

2.1.2 Class 2: Stranded Conductors

Consists of multiple thick strands.

- **Properties:** Rigid enough to be self-supporting on trays but allows for some bending.
- **Application:** Standard for **Steel Wire Armoured (SWA)** cables.

2.1.3 Class 5: Flexible Conductors (Preferred)

Constructed from numerous fine wires (typically 0.25mm to 0.4mm diameter).

- **Properties:** High flexibility while retaining mechanical strength.
- **Application:** The standard for **H07RN-F** cables.
- **UCL Preference:** This is the primary conductor type for our mobile lab equipment and fixed installations subject to vibration or complex routing.

2.2 Insulation Materials

Table 2.1: Comparison of Insulation Materials

Material	Properties	Max Temp
PVC (Thermoplastic)	Standard insulation for SWA. Softens significantly above rated temp.	70°C
EPR (Thermosetting)	Used in H07RN-F . Excellent dielectric strength. Resists thermal shock up to 250°C during faults.	90°C
XLPE (Thermosetting)	Modern SWA insulation. High thermal performance.	90°C

H07RN-F Breakdown:

- **H:** Harmonised Standard.
- **07:** 450/750V Voltage Rating.
- **R:** EPR Insulation.
- **N:** PCP (Polychloroprene/Neoprene) Sheath – Oil, weather, and abrasion resistant.
- **F:** Flexible Copper Strands (Class 5).

Chapter 3

Cable Selection: H07RN-F vs. SWA

Our primary preference is H07RN-F. However, regulatory compliance requires specific conditions to be met when using flexible cables in fixed installations.

3.1 Flexible Cables in Fixed Installations

Regulation 521.9.1 (BS 7671) states that flexible cables shall be used for fixed wiring only where the relevant provisions of the Regulations are met. They must be of the heavy-duty type unless mechanical damage risk is low.

H07RN-F Compliance: H07RN-F is classified as heavy-duty. It is permissible for fixed installations provided:

1. **Mechanical Protection:** If installed in areas subject to impact (e.g., low-level workshop walls), it must be protected by conduit, trunking, or located on a tray where impact risk is mitigated.
2. **Termination:** Class 5 conductors must be terminated with ferrules (see Chapter 6).

3.2 Selection Decision Matrix

Table 3.1: Cable Selection Matrix

Feature	H07RN-F (Preferred)	SWA (Alternative)
Flexibility	High (Dynamic/Coiling)	Low (Static Only)
Vibration Tolerance	Excellent (Absorbs vibration)	Poor (Gland fatigue)
Water Resistance	AD8 (Submersible)	AD7 (With correct gland)
Mechanical Crush	Medium (Rubber Sheath)	High (Steel Armour)
Direct Burial	No (Requires Duct)	Yes

When to use SWA:

- Direct burial in ground (without ducting).
- Areas of high mechanical risk where additional containment (trunking/conduit) is not feasible.

Chapter 4

Design and Sizing Methodology

4.1 The Coordination of Characteristics

The design must satisfy the fundamental inequality:

$$I_b \leq I_n \leq I_z \quad (4.1)$$

Where:

- I_b : Design Current (The load).
- I_n : Protection Rating (Fuse/Breaker).
- I_z : Current Carrying Capacity of the cable in installed condition.

4.2 Calculating Tabulated Current (I_t)

To determine the required cable size, calculate the minimum tabulated current (I_t) required before derating:

$$I_t \geq \frac{I_n}{C_a \times C_g \times C_i} \quad (4.2)$$

4.2.1 Ambient Temperature Factor (C_a)

For Thermosetting (EPR/H07RN-F) cables rated at 90°C:

Temp (°C)	30	40	50	60	70
Factor (C_a)	1.00	0.91	0.82	0.71	0.58

4.2.2 Grouping Factor (C_g) - Table 4C1

Avoid grouping where possible. If cables touch, apply:

- **2 Circuits:** 0.80
- **3 Circuits:** 0.70
- **4 Circuits:** 0.65
- **6 Circuits:** 0.57

4.3 H07RN-F Current Ratings

Derived from BS 7671 Table 4F1A. Ambient 30° C, Conductor 60° C.

Table 4.1: Current Carrying Capacity (Amps)		
Size (mm ²)	Single Phase (1-Ph)	Three Phase (3-Ph)
1.0	10	10
1.5	16	16
2.5	25	20
4.0	32	30
6.0	43	38
10.0	60	53
16.0	79	71
25.0	104	94

4.4 Voltage Drop

Max permitted drop: **3% Lighting, 5% Power.**

$$V_d = \frac{(mV/A/m) \times I_b \times L}{1000} \quad (4.3)$$

H07RN-F Voltage Drop Data (mV/A/m):

- **1.5mm²**: 32 (1-Ph) / 27 (3-Ph)
- **2.5mm²**: 19 (1-Ph) / 16 (3-Ph)
- **4.0mm²**: 12 (1-Ph) / 10 (3-Ph)
- **6.0mm²**: 7.8 (1-Ph) / 6.7 (3-Ph)

Chapter 5

Installation and Mechanical Protection

5.1 Clipping and Support Spacing

Flexible cables used in fixed installations must be supported to prevent strain on terminations.

Table 5.1: Recommended Support Spacing (H07RN-F)

Cable Diameter (mm)	Horizontal (mm)	Vertical (mm)
< 9	250	400
9 - 15	300	400
15 - 20	350	450
20 - 40	400	550

Note on Escape Routes: In accordance with BS 7671 Regulation 521.10.202, cables in escape routes must be supported by fire-resistant (metal) fixings to prevent premature collapse. Plastic clips alone are prohibited.

5.2 Bending Radius

To prevent damage to the internal structure:

- **Fixed Installation:** Minimum $4 \times$ Overall Diameter.
- **Flexing Application:** Minimum $6 \times$ Overall Diameter.

5.3 Mechanical Protection Strategies

Where H07RN-F is used in areas of risk:

1. **Conduit:** Use heavy-gauge or flexible conduit (Kopex/Adaptaflex) for the section of cable exposed to damage.
2. **Tray/Basket:** Run cables at a high level ($>2\text{m}$) on the cable tray.
3. **Positioning:** Route cables within the structural voids of machinery frames.

5.4 Hazardous Environments (DSEAR & Environmental Extremes)

Laboratories pose unique environmental challenges that supersede standard commercial installation methods. When routing cables through these areas, the following mitigations are mandatory:

- **Wet Labs & Fluid Testing:** All cables routed through wet zones or areas subject to wash-down must utilize IP67 or IP68 rated compression glands (Nylon or Nickel-Plated Brass).
- **Explosive Atmospheres (DSEAR):** In areas where battery off-gassing or hydrogen use creates a zoned explosive atmosphere, standard H07RN-F glands are strictly prohibited. ATEX-certified (Ex d or Ex e) barrier glands must be used to prevent cables from acting as conduits for explosive gases.
- **Chemical Degradation:** While H07RN-F features a PCP (Neoprene) sheath resistant to general oil and weathering, prolonged exposure to concentrated solvents (e.g., in cleanroom environments) requires the substitution of standard cables with PTFE or specialized chemical-resistant sheathing.

Chapter 6

Termination and Connectivity

6.1 Termination of Class 5 Conductors

Critical Rule: Direct termination of fine stranded Class 5 conductors into screw terminals is prohibited as the screw shears the strands.

6.1.1 Bootlace Ferrules

All Class 5 terminations must use insulated bootlace ferrules crimped with a ratchet tool to ensure a gas-tight joint.

Table 6.1: Ferrule Colour Codes (DIN 46228 Part 4)

CSA (mm ²)	French/DIN Colour	German Colour
0.5	White	Orange
0.75	Blue	White
1.0	Red	Yellow
1.5	Black	Red
2.5	Grey	Blue
4.0	Orange	Grey
6.0	Green	Black
10.0	Brown	Ivory

6.2 Gland Selection

6.2.1 For Unarmoured (H07RN-F)

Use Compression Glands (Nylon or Nickel-Plated Brass).

- **IP Rating:** Must maintain IP68 sealing against the outer sheath.
- **Strain Relief:** The gland must secure the cable such that no tension is transferred to the conductor terminations.

Table 6.2: Metric Compression Gland Sizes

Thread	Cable Range (mm)
M20	6 - 12
M25	11 - 17
M32	15 - 21
M40	19 - 28

6.2.2 For Armoured (SWA) - If Necessary

- **Internal (Dry):** BW Gland (2-part).
- **External (Wet):** CW Gland (3-part with outer seal).
- **Earthing:** A "Banjo" earth tag and fly-lead to the earth terminal are mandatory.

Chapter 7

Plug Selection (BS EN 60309)

Industrial plugs must be selected to prevent cross-connection of different voltages.

7.1 Color Coding and Voltage

- **Yellow:** 110V AC (Construction/Safety).
- **Blue:** 230V AC (Standard Single Phase).
- **Red:** 400V AC (Three Phase).
- **Black:** 500V - 690V AC.

7.2 Pin Configuration (Clock Position)

The earth pin is located at a specific "hour" position relative to the keyway (6h) to mechanically prevent mismatching.

Table 7.1: BS EN 60309 Earth Pin Positions

Voltage	Pins	Earth Pos	Colour
110V	2P + E	4h	Yellow
230V	2P + E	6h	Blue
400V	3P + E	6h	Red
400V	3P + N + E	6h	Red

7.3 IP Ratings

- **IP44 (Splashproof):** Spring-loaded flap. Suitable for indoor workshops.
- **IP67 (Watertight):** Twist-lock ring. **Mandatory** for outdoor extension leads or wash-down areas.

Chapter 8

Verification and Testing

Before energizing any new cable run, the following tests are mandatory under BS 7671.

8.1 Continuity of CPC (R_2)

Verify the integrity of the earth path.

- For H07RN-F: Test the Green/Yellow core.
- For SWA: Test the steel armour (via the gland banjo).

8.2 Insulation Resistance (IR)

Verify no short circuits or damage to insulation.

- **Test Voltage:** 500V DC.
- **Minimum Result:** $> 1.0M\Omega$ (BS 7671 limit).
- **Expected Result:** $> 200M\Omega$ (New cable).

8.3 Earth Loop Impedance (Z_s)

Ensure the protective device will disconnect within the required time (0.4s for TN systems).

$$Z_s = Z_e + (R_1 + R_2) \quad (8.1)$$

The measured Z_s must be less than the maximum Z_s values in BS 7671 Tables 41.2/41.3.

8.4 Information Provision and App Integration

To ensure top-down management visibility and compliance with the information provision requirements of PUWER, all verification test results (R_2 , IR, and Z_s) must be recorded.

Table 8.1: Max Z_s for Type B and C MCBs (BS 7671 Tables 41.2/41.3)

Device Rating (I_n)	Type B ($Z_s \Omega$)	Type C ($Z_s \Omega$)
6 A	7.28	3.64
10 A	4.37	2.19
16 A	2.73	1.37
20 A	2.19	1.09
25 A	1.75	0.87
32 A	1.37	0.68
40 A	1.09	0.55
50 A	0.87	0.44
63 A	0.69	0.35

Note: Values assume U_0 of 230V and incorporate the C_{min} factor.